

Learning Dynamics & Catastrophic Forgetting in Small Language Model

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Codebase Link: <https://github.com/rituraj-sharma/lmcf>

Models Link:

https://drive.google.com/drive/folders/1_qNfhsBrWCed6_IAMgFkcjkZGTKsweO5?usp=sharing

Introduction

Large language models are typically trained on static, heterogeneous corpora where all data is available simultaneously. In many real-world scenarios, data arrives sequentially, a model trained on one domain may later need to acquire knowledge from another without access to the original training data. This setting, known as continual learning, introduces the well-documented problem of catastrophic forgetting, where the acquisition of new knowledge interferes with and degrades previously learned representations.

In this project, we investigate continual learning in the context of two linguistically distinct domains:

1. **Domain A:** Children Book Stories (Dataset: TinyStories) and
2. **Domain B:** News Articles (Dataset: AG News)

across three model scales. This contrast allows us to study not only whether forgetting occurs, but whether its magnitude and distribution within the network depend on the structural properties of the domains involved and the order in which they are encountered.

We conduct a controlled set of experiments with the following research questions:

RQ1 — Does catastrophic forgetting occur under sequential domain training, and how severe is it?

We evaluate the degree to which a model trained first on one domain and subsequently on another retains knowledge of the original domain, using perplexity, backward transfer, and retention scores.

RQ2 — Does training order affect the degree and pattern of forgetting? We compare two curriculum orderings: sequential (stories → news) and reverse (news → stories) to determine whether the direction of domain transition influences how much is forgotten and whether forgetting is symmetric across domains.

RQ3 — Does model scale modulate resistance to forgetting? We train models at small, medium, and large scales under identical conditions to assess whether larger models are inherently more robust to catastrophic forgetting or merely delay its onset.

Ablation Design

To systematically address the above research questions, we design the following ablation conditions:

- **Mixed training (E2 - baseline):** All models are trained jointly on a combined stories and news corpus, establishing an upper-bound reference for both domains simultaneously.
- **Sequential ordering (E1):** Models are trained first on the stories domain (Stage 1) and subsequently on the news domain (Stage 2), representing a simple-to-complex curriculum.
- **Reverse ordering (E3):** Models are trained first on the news domain (Stage 1) and subsequently on the stories domain (Stage 2), representing a complex-to-simple curriculum.
- **Model scale:** Each training condition is run independently at three parameter scales: M1 (Small), M2 (Medium), and M3 (Large) to isolate the effect of capacity on forgetting and transfer.
- **Evaluation stage:** Performance is recorded at the end of both Stage 1 and Stage 2 for all continual learning conditions, enabling direct measurement of knowledge degradation between stages.

Experimental Methodology:

This study investigates catastrophic forgetting and curriculum learning in small Transformer language models under sequential domain adaptation. We train decoder-only autoregressive language models across two linguistically distinct domains: Domain A (simple children's narratives) and Domain B (structured news text), following three distinct training regimes. All experiments are conducted at three model scales to examine whether parameter count influences the degree of forgetting, knowledge retention, and forward transfer.

The experimental design follows a strict controlled-variable approach: the tokenizer, data budget, sequence length, and training procedure are held constant across all model scales. Only the model architecture (number of layers, hidden dimension) varies. This ensures that any observed differences in forgetting scores across M1, M2, and M3 are attributable to model capacity rather than data or training confounds.

Datasets

Two publicly available datasets were selected to represent maximally contrasting linguistic domains. Domain A (TinyStories) consists of synthetically generated children's stories using a simple vocabulary, whereas Domain B (AG News) comprises four-class news articles with complex vocabulary and diverse topic coverage. The structural asymmetry between domains, particularly the difference in average text length, is acknowledged as a design characteristic that reflects realistic domain shift conditions.

Property	Domain A — TinyStories	Domain B — AG News
Source	roneneldan/TinyStories (HuggingFace)	fancyzhx/ag_news (HuggingFace)
Total texts	~2.1 million stories	120,000 articles (4 classes)

Property	Domain A — TinyStories	Domain B — AG News
Token budget (M1)	2M BPE tokens	2M BPE tokens
Token budget (M2)	4M BPE tokens	4M BPE tokens
Token budget (M3)	6M BPE tokens	6M BPE tokens
Avg tokens / text	~222 tokens	~52 tokens
Vocabulary style	Simple, repetitive, ~1,500 core words	Complex, diverse, news-specific
Train / Val / Test	70% / 15% / 15%	70% / 15% / 15%

Table 1. Dataset statistics. Token budgets are BPE-tokenized counts at $MAX_SEQ_LEN=128$.

A shared BPE tokenizer (vocab_size=8,000) was trained once on the combined corpus of both domains and fixed across all model scales and experiments. This is critical for experimental validity: changing the tokenizer with scale would confound the measurement of forgetting by simultaneously altering sequence lengths, prediction difficulty, and domain overlap — making it impossible to isolate the effect of model capacity.

Token budgets are intentionally held below the AG News maximum capacity (~6.25M tokens) and scaled with model size (2M/4M/6M for M1/M2/M3) to reflect realistic compute budgets.

Model Configurations

Three decoder-only Transformer language models of increasing scale were designed to span a parameter range representative of small LMs. All three share the same architectural template: causal self-attention with FlashAttention (scaled dot-product attention with is_causal=True), pre-norm layer normalisation, GELU feedforward networks, and tied input/output embeddings. The feed-forward dimension is fixed at $4 \times d_{\text{model}}$ for all scales, consistent with standard transformer design.

Scale	Params	Layers	d_{model}	Heads	FFN Dim	LR	Dropout
M1 (Small)	~1.5M	2	128	4	512	3×10^{-4}	0.1
M2 (Medium)	~7M	6	256	8	1024	2×10^{-4}	0.1
M3 (Large)	~24M	12	384	12	2048	1×10^{-4}	0.1

Table 2. Model configurations. $d_k = d_{\text{model}} / \text{heads} = 32$ (M1/M2) or 64 (M3). All models use weight tying between token embedding and lm_head .

Learning rates decrease with scale following standard practice as larger models are more sensitive to gradient updates and require smaller step sizes for stable training. Dropout (0.1) is same for M1, M2 and M3.

Common Training Configuration

The following hyperparameters and procedures are identical across all model scales and experiments. This design ensures that any observed differences between M1, M2, and M3 are attributable solely to model capacity.

Parameter	Value	Rationale
Sequence length (MAX_SEQ_LEN)	128 tokens	Fixed context window across all scales
Batch size	256 sequences	~32,768 tokens/batch
Tokenizer	BPE, vocab_size=8,000	Trained once on A+B corpus, shared across scales
Optimiser	AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\lambda=0.1$)	Standard for LM pre-training
LR schedule	Linear warmup (10%) + cosine decay to $lr/100$	Stage 2 resets LR from scratch, clean gradient momentum
Gradient clipping	max norm = 1.0	Prevents gradient explosion in early training
Mixed precision	AMP float16 (GPU)	Reduces memory and speeds training
Early stopping patience	5 eval intervals	Stops if val PPL does not improve for 1,000 steps
Eval interval (in-training)	Every 200 steps (subset: 10 batches)	Fast approximation for convergence signal
Eval at stage boundary	Full val + test set	Accurate final metrics from best checkpoint
Data split	70 / 15 / 15 (train/val/test)	Test set used only for final evaluation

Table 3. Common training configuration shared across all model scales and experiments.

Experimental Design

Three experiments are conducted for each model scale, yielding 9 training runs in total (3 scales $\times$ 3 experiments). Each experiment consists of one or two sequential training stages. Stage 2 always begins from Stage 1's best checkpoint (determined by lowest validation perplexity) rather than from the final epoch - this ensures Stage 2 starts from the most well-consolidated representation of Stage 1's domain.

Exp.	Order	Stage 1	Stage 2	Primary Metric
E1	A $\rightarrow$ B	Train on Domain A from scratch	Train on Domain B from S1 best	BWT_A: forgetting of A after learning B
E2	Mixed	Train on A+B simultaneously	— (single stage)	Joint baseline: no forgetting by design
E3	B $\rightarrow$ A	Train on Domain B from scratch	Train on Domain A from S1 best	BWT_B: forgetting of B after learning A

Table 4. Experimental design. E2 serves as the joint training baseline. Each experiment run $\times$ 3 model scales = 9 total runs.

E1 and E3 are designed symmetrically to test whether forgetting is domain-directional: E1 measures how much Domain A (simple, structured) is forgotten after learning Domain B (complex, diverse), while E3 measures the reverse. E2 provides the mixed training baseline representing the theoretical upper bound on joint performance, a model that never experiences catastrophic forgetting because both domains are presented simultaneously.

Evaluation Metrics

Perplexity (PPL)

Perplexity is the primary evaluation signal, computed as $\exp(\text{cross-entropy loss})$ on held-out val and test sets. PPL on both domains is measured after every stage from the best checkpoint, not the final training step, ensuring metrics reflect the model's optimal learned state rather than potential overfitting at training end.

Backward Transfer (BWT) and Retention

BWT quantifies catastrophic forgetting as the increase in perplexity on the original domain after learning a new one. Both raw (PPL points) and normalised (fraction of baseline) versions are reported. Retention Rate ($S1_PPL / S2_PPL$) provides an intuitive complement, where 1.0 indicates perfect retention and lower values indicate greater forgetting.

Metric	Formula	Interpretation
BWT_raw	$PPL_A(S2) - PPL_A(S1)$	Positive = forgetting occurred
BWT_norm	$(PPL_A(S2) - PPL_A(S1)) / PPL_A(S1)$	Scale-independent; use for cross-model comparison
Retention	$PPL_A(S1) / PPL_A(S2)$	1.0 = perfect retention; 0 = total forgetting
FWT_raw	$PPL_B(E1_S2) - PPL_B(E3_S1)$	Negative = A pretraining helped learn B
FWT_norm	$(PPL_B(E1_S2) - PPL_B(E3_S1)) / PPL_B(E3_S1)$	Normalised forward transfer
Mixed Gap	$PPL_A(E1_S2) - PPL_A(E2)$	Cost of sequential vs joint training

Table 5. Evaluation metrics. All metrics computed on validation set; test set used for final reported values only.

Semantic Metrics

To capture domain forgetting beyond token-level statistics, two semantic evaluation metrics are computed from 1,000 generated samples (500 per domain prompt) using the best checkpoint at each stage. Concept Coverage measures the fraction of domain-characteristic vocabulary present in generated text, where domain concepts are automatically extracted via TF-IDF differential scoring. BERTScore computes token-level semantic similarity between Stage 1 and Stage 2 generations on identical prompts: a drop in F1 indicates semantic domain forgetting beyond what perplexity captures.

Results Analysis:

Baseline: Mixed Domain Training:

We first establish a multi-domain baseline by training all three model scales jointly on a mixed corpus of stories and news text. The mixed setting represents the ideal upper bound where no ordering constraint exists.

Table 1. Model Scale wise baseline (mixed) score

Model Scale	PPL _{mixed}	PPL _{stories}		PPL _{news}	
	Val	Val	Test	Val	Test
M1 (Small)	66.190	25.35	26.04	150.30	152.56
M2 (Medium)	22.41	17.07	16.82	100.66	103.14
M3 (Large)	14.58	16.86	16.73	105.47	104.93

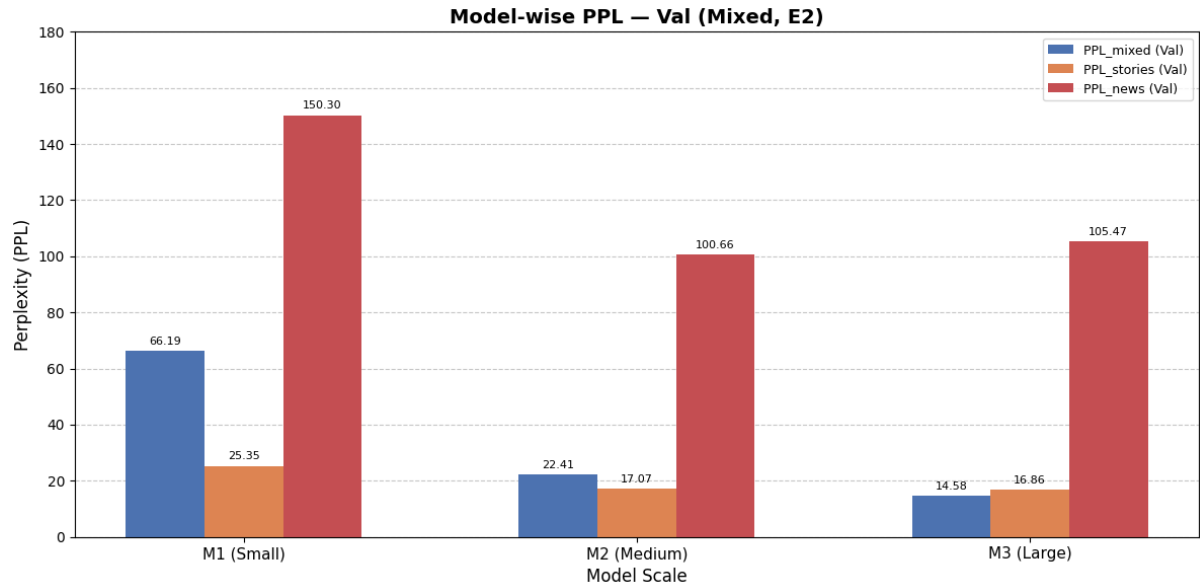


Figure 1 Model Scale wise PPL Score for Baseline

Continual Learning: Sequential and Reverse Ordering

Table 2 presents perplexity results across both stages of training for sequential and reverse curriculum orderings. The results reveal a strong and consistent pattern of catastrophic forgetting under both orderings, with the direction of forgetting determined by the training order.

Table 2 Model Scale wise PPL Score for Sequential and Reverse Learnings

Model Scale	Order	Stage 1 (S1)				Stage 2 (S2)			
		PPL _{stories}		PPL _{news}		PPL _{stories}		PPL _{news}	
		Val	Test	Val	Test	Val	Test	Val	Test
M1 (Small)	Sequential	19.74	20.21	22708.34	22402.25	413.59	418.14	92.70	93.89
	Reverse	1040.79	1050.38	115.54	117.33	16.08	16.41	5898.57	5820.31
M2 (Medium)	Sequential	12.73	12.43	9606.72	9574.53	259.48	259.21	53.41	54.55
	Reverse	814.41	824.22	73.06	74.86	10.28	10.04	7014.82	7000.86
M3 (Large)	Sequential	12.18	12.10	6956.10	6950.64	195.15	196.86	50.45	50.16
	Reverse	744.51	754.30	76.08	75.53	9.56	9.53	2884.44	2878.10

Sequential ordering (stories → news): Models achieve strong Stage 1 performance on the stories domain, with M1, M2, and M3 reaching val PPLs of 19.74, 12.73, and 12.18 respectively, all close to the mixed-training baseline. However, after Stage 2 news training, stories PPL degrades dramatically to 413.59, 259.48, and 195.15 for M1, M2, and M3, representing a severe loss of stories knowledge. News domain PPL after Stage 2 is comparatively healthy (92.70, 53.41, and 50.45), indicating that the model successfully learns the second domain at the expense of the first.

Reverse ordering (news → stories): The mirror pattern emerges. Stage 1 news PPL is extremely high for sequential-order models (M1: 22,708.34; M2: 9,606.72; M3: 6,956.10), reflecting the difficulty of learning news as the initial domain. After Stage 2 stories training, news PPL collapses further to 5,898.57, 7,014.82, and 2,884.44 for M1, M2, and M3 respectively, confirming catastrophic forgetting of the first-trained domain in both orderings. Stories PPL at Stage 2 is well-recovered (16.08, 10.28, 9.56), again suggesting that the second-stage domain dominates the model's knowledge.

Across both orderings, larger models tend to exhibit lower absolute PPL values but are not immune to forgetting. The relative degradation remains substantial at all scales.

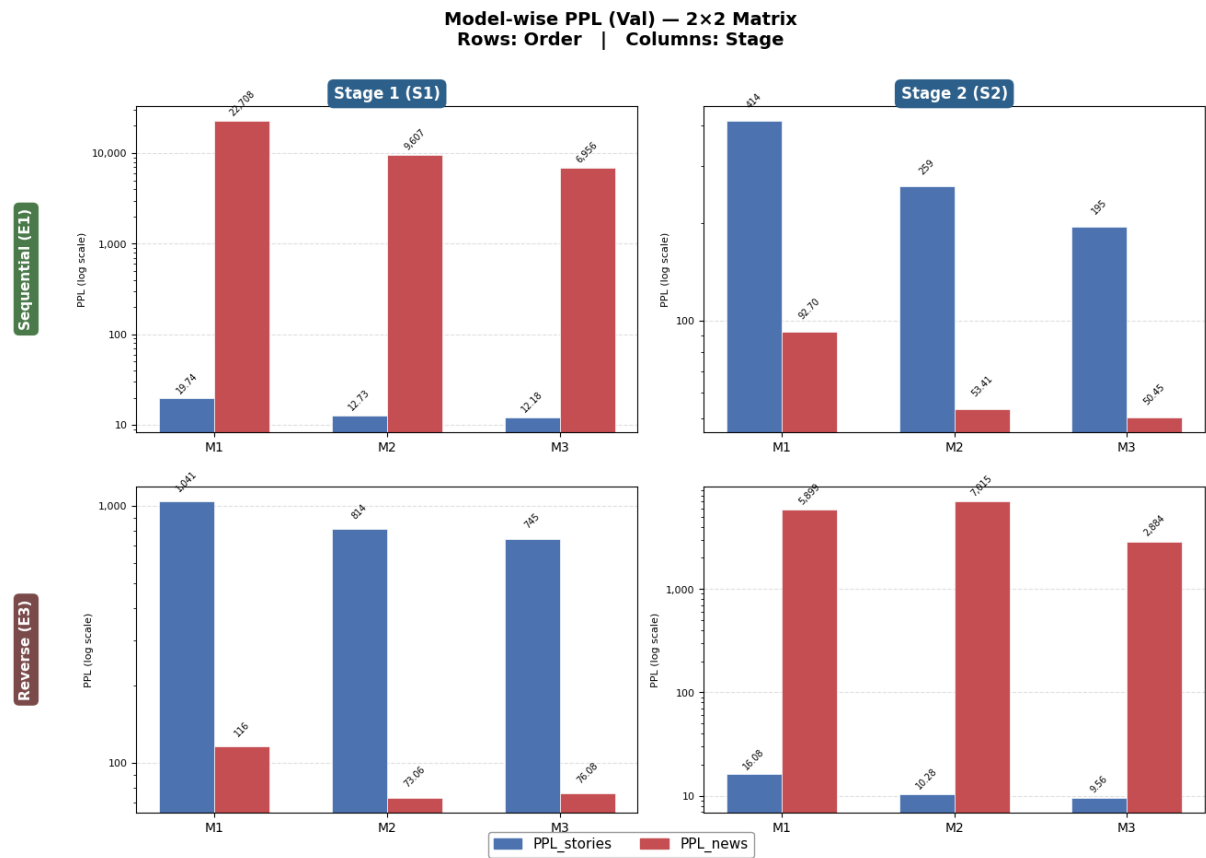


Figure 2 Model wise PPL comparison between Sequential and Reverse order across Stages

Forgetting and Retention Analysis

Table 3 quantifies forgetting through backward transfer (BWT), retention, and gap scores. Under sequential ordering, BWT for the stories domain is +393.85, +246.74, and +182.96 for M1, M2, and M3, corresponding to roughly 20×, 19×, and 15× degradation relative to baseline. The news domain suffers even more severely, with BWT values of +5,783.04, +6,941.76, and +2,808.35, equivalent to 50×, 95×, and 37× forgetting. Retention scores are near zero across all models and domains (ranging from 0.01 to 0.06), confirming that almost no knowledge of the first-trained domain is preserved after the second stage. Gap scores, which measure the distance between current performance and baseline, are consistently in the hundreds to thousands of PPL points, highlighting the magnitude of the forgetting problem.

Table 3 Forgetting Score across model scale and order of learnings

Model Scale	Sequential (E1)			Reverse (E2)		
	BWT _{stories} (Norm)	Ret _{stories}	Gap _{stories}	BWT _{news} (Norm)	Ret _{news}	Gap _{news}
M1 (Small)	+393.85 (+20x)	0.04	+388.24	+5783.04 (+50x)	0.02	+5748.27
M2 (Medium)	+246.74 (+19x)	0.05	+242.41	+6941.76 (+95x)	0.01	+6914.15
M3 (Large)	+182.96 (+15x)	0.06	+178.28	+2808.35 (+37x)	0.03	+2778.97

Backward Transfer (Forgetting) by Model Scale

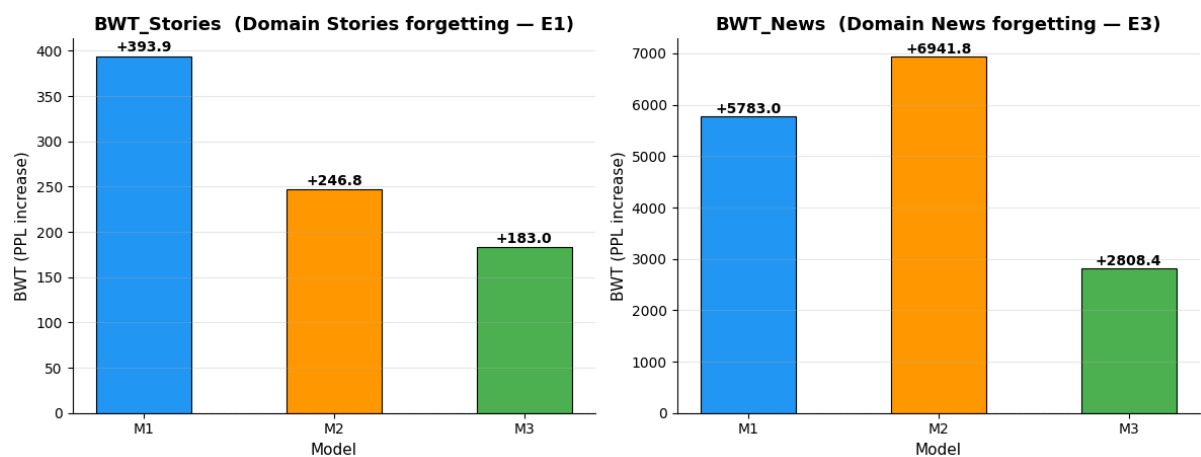
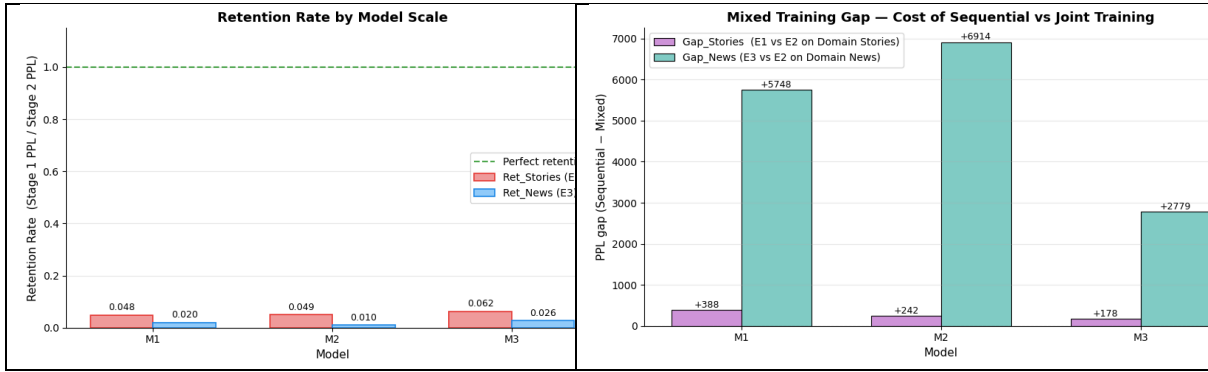


Figure 3 Backward Transfer Score (Forgetting by Model Scale)



Asymmetry in Forgetting. A key finding, illustrated in Figure 4, is the asymmetry in forgetting between domains. News is consistently forgotten at a much higher rate than stories regardless of model scale, with M2 exhibiting the huge divergence (95× vs. 19×). This asymmetry reflects the structural difference between the two domains: the narrow, formulaic register of news text is far more easily overwritten by the richer, more diverse stories distribution than vice versa. Crucially, this asymmetry persists across all three model scales and narrows only marginally with increasing capacity.

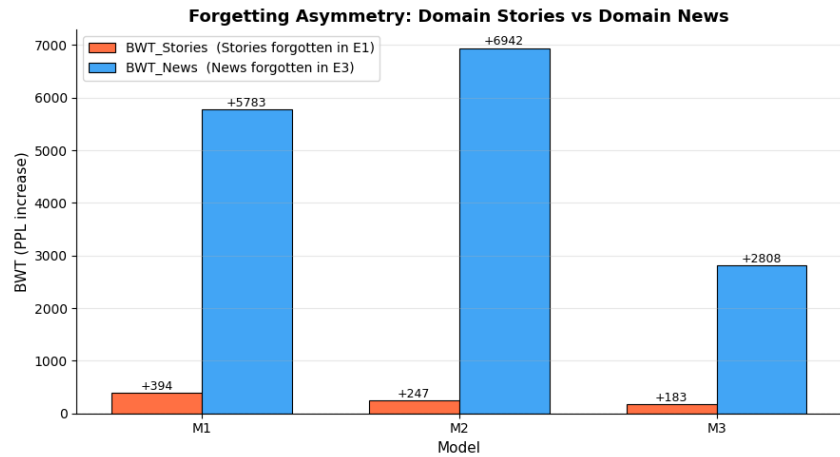


Figure 4 Forgetting Asymmetry across domain

M3 (Large) shows the smallest absolute forgetting values overall, suggesting larger models offer modest resistance to forgetting, though near-zero retention at all scales confirms that model size alone cannot resolve catastrophic forgetting under strict sequential training.

Forward Transfer

Table 4 reports forward transfer (FWT) scores, which measure whether training on one domain beneficially accelerates learning on the subsequent domain. Negative FWT values indicate positive transfer. All models exhibit negative FWT for both domains, with FWT_stories ranging from -2.44 to -3.66 (approximately 0.18 – $0.21\times$ improvement) and FWT_news ranging from -19.65 to -25.62 (approximately 0.20 – $0.34\times$ improvement). This indicates that pre-exposure to one text domain provides a modest but consistent learning advantage when the model is subsequently trained on the other domain. The news domain benefits more from stories pre-training, likely because the stories corpus provides richer linguistic priors. Forward transfer improves marginally with model scale for the news domain ($0.20\times$ for M1 to $0.34\times$ for M3), suggesting that larger models better leverage prior domain knowledge when learning a new one.

Table 4 Forward Transfer Score across Model Scale

Model Scale	FWT _{stories} (Norm)	FWT _{news} (Norm)
M1 (Small)	-3.66 (0.18x)	-22.84 (0.20x)
M2 (Medium)	-2.44 (0.19x)	-19.65 (0.27x)
M3 (Large)	-2.62 (0.21x)	-25.62 (0.34x)

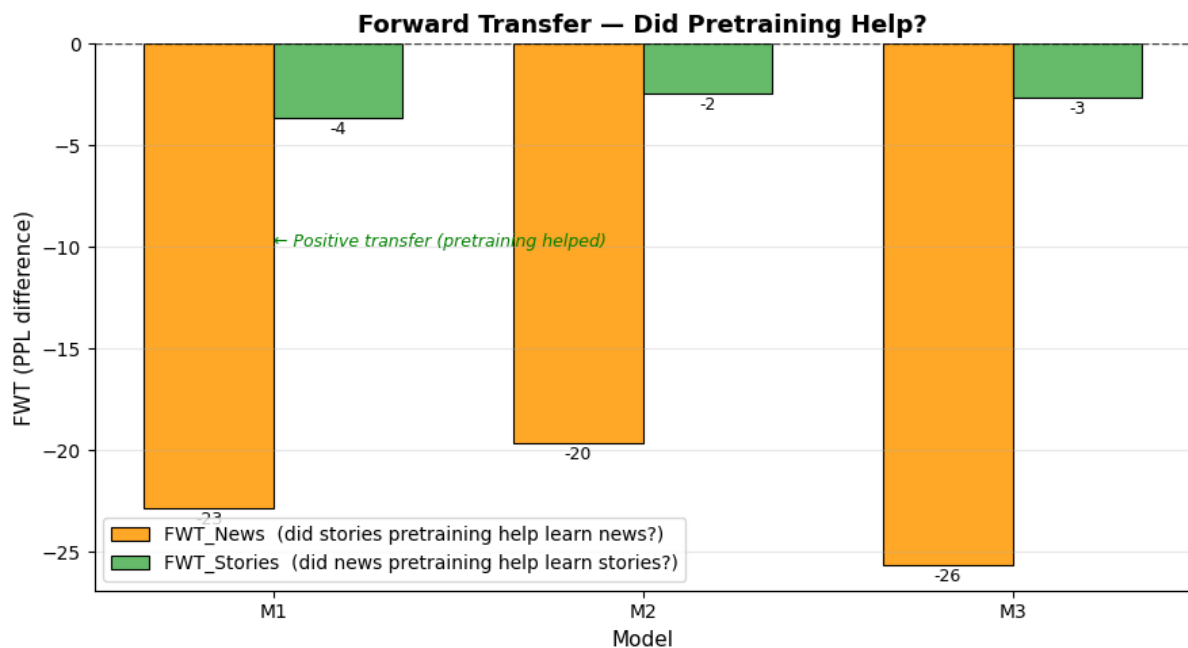


Figure 5 Forward Transfer Score comparison across models and domains

Semantic Quality: BERTScore and Coverage

We examine the semantic dimension of continual learning through BERTScore F1 and Coverage Score (Table 5) under sequential and reverse orderings.

BERTScore F1 remains remarkably stable across all models, orderings, and prompt types, with values tightly clustered between 0.800 and 0.810. Neither sequential nor reverse ordering leads to noticeable degradation in surface-level semantic similarity. This suggests that while PPL-based metrics reveal catastrophic forgetting at the distributional level, the models retain sufficient linguistic competence to generate semantically coherent text across both domains regardless of training order.

Table 5 BERTScore F1 and Coverage Score across model scale and domain

Model Scale	Order	BERTScore F1		Coverage Score			
		Prompt _{stories}	Prompt _{news}	Prompt _{stories}		Prompt _{news}	
				Domain _{stories}	Domain _{news}	Domain _{stories}	Domain _{news}
M1 (Small)	Sequential	0.807	0.8005	0.54	0.88	0.26	0.86
	Reverse	0.8044	0.8003	0.98	0.02	0.98	0.08
M2 (Medium)	Sequential	0.8088	0.8029	0.54	0.88	0.48	0.88
	Reverse	0.8059	0.8033	1	0.02	1	0.08
M3 (Large)	Sequential	0.8098	0.8043	0.58	0.88	0.42	0.9
	Reverse	0.8046	0.8021	1	0.06	0.98	0.06

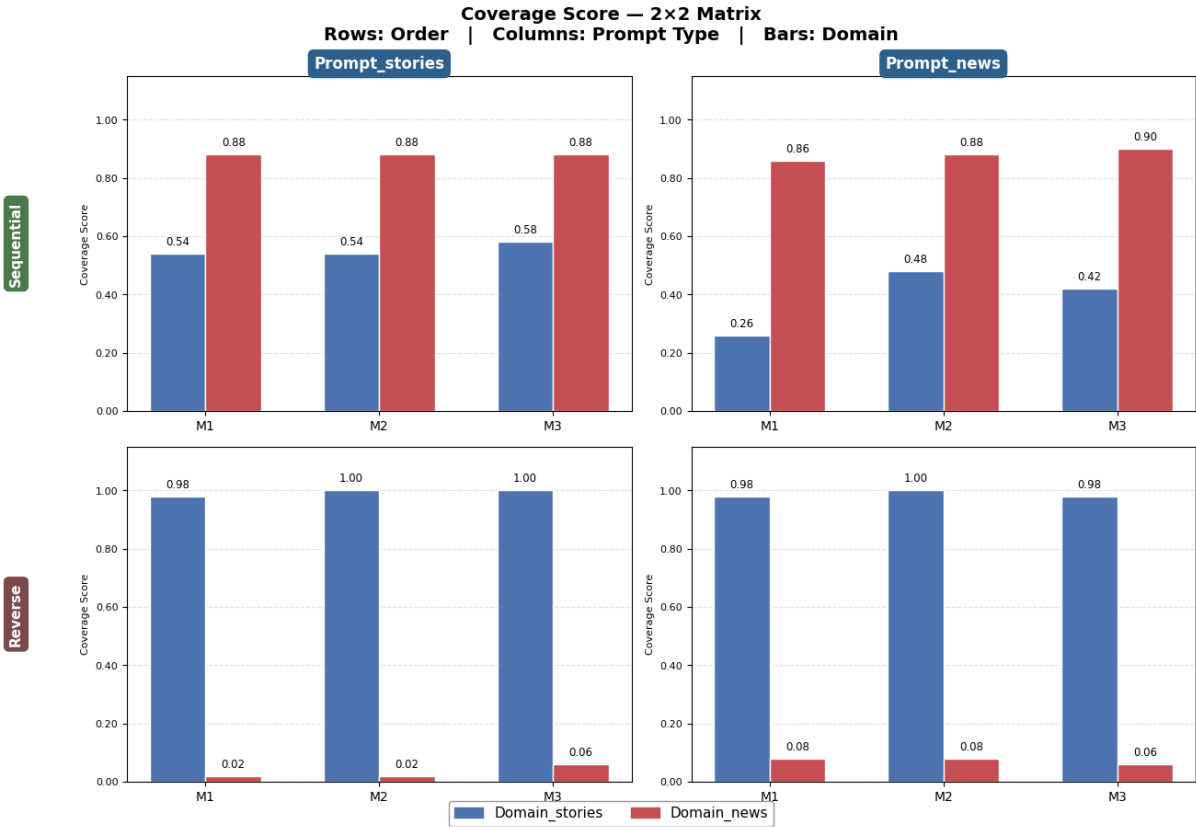


Figure 6 2x2 Grid Matrix: Coverage Score comparisons across orders

Coverage Score, however, tells a more nuanced story. Under sequential ordering, prompts drawn from the stories domain show moderate domain_stories coverage (0.54–0.58) alongside high domain_news coverage (0.88), indicating that even when prompted with stories-style text, the model's outputs lean toward the most recently trained (news) domain. Under reverse ordering, this pattern flips sharply: stories-prompted outputs achieve near-perfect domain_stories coverage (0.98–

1.0) but almost zero domain_news coverage (0.02–0.06), confirming that the model's generation is dominated by the second-stage training domain.

The coverage pattern for Prompt_news mirrors these findings. Sequential models retain strong news coverage after Stage 2 news training, while reverse models produce news-prompted outputs that almost entirely reflect the stories distribution after Stage 2. Across model scales, M3 shows slightly more balanced coverage values in several conditions, consistent with its greater capacity, though the directional pattern of domain dominance is consistent across all scales.

Taken together, the semantic metrics reveal that catastrophic forgetting manifests not only as increased perplexity but also as a systematic shift in the model's generative distribution toward the most recently trained domain, a qualitative change that BERTScore alone is insufficient to capture, while coverage scores directly expose it.

Weight change by layer type:

To better understand where forgetting occurs mechanistically, Figure 7 examines where forgetting occurs by plotting the average relative weight change from Stage 1 to Stage 2, broken down by layer type for sequential (E1) and reverse (E3) orderings. Notably, the y-axis scales differ between the two plots, E3 reaches up to 1.4 compared to 0.7 for E1, indicating that the reverse ordering induces larger overall weight perturbations.

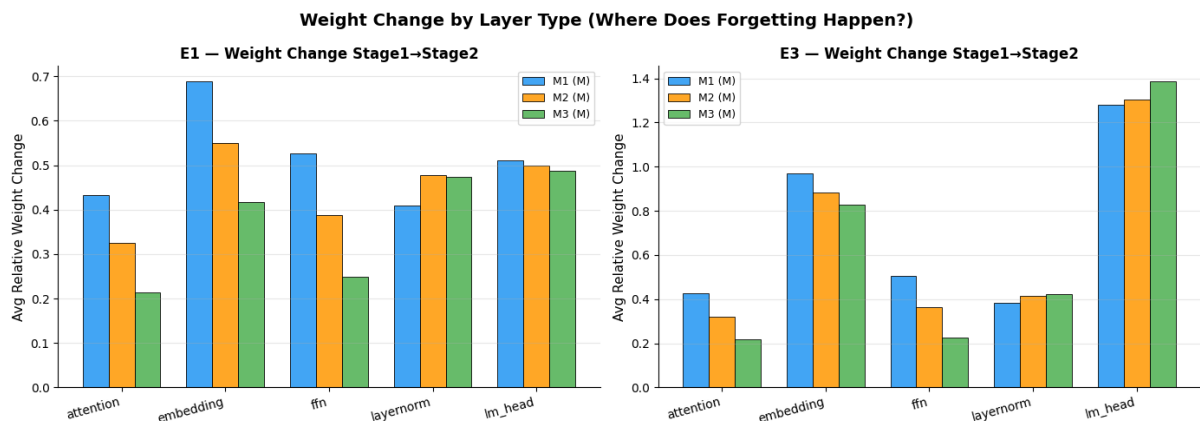


Figure 7 Weight change by layer across order and model scale

Effect of Model Scale. Across both orderings, M1 (Small) consistently shows the largest weight changes while M3 (Large) shows the smallest, holding across all layer types. This confirms that larger models possess greater representational stability, requiring less aggressive overwriting to accommodate new domain data.

Sequential Ordering (E1 — Simple to Complex). Weight changes are distributed relatively uniformly across all layer types: attention, embedding, FFN, layer norm, and lm_head all shift by comparable magnitudes. Training from a simpler to a more complex domain therefore demands broad network-wide reorganisation, as the model cannot rely on any particular subset of layers to absorb the new domain and must adapt at every level.

Reverse Ordering (E3 — Complex to Simple). E3 tells a different story. The lm_head dominates with changes reaching 1.2–1.4, while attention layers remain comparatively stable. Having already learned the complex news domain, the model retains its deep representational structure and concentrates adaptation in the output classifier, which recalibrates the generation distribution toward the simpler stories domain without restructuring internal representations.

Together, these findings show that the direction of domain transition determines not just the magnitude but the location of weight change — a distinction relevant for continual learning strategies that selectively freeze or regularise specific layer types.

Key Observations

1. **Scale improves resistance to forgetting but does not eliminate it.** Larger models consistently achieve lower perplexity and smaller weight shifts across all layer types, indicating greater representational stability and stronger resistance to knowledge overwriting. However, near-zero retention scores across all scales confirm that increased model capacity alone is insufficient to prevent catastrophic forgetting under sequential training.
2. **Forgetting is severe under both orderings, but training direction matters.** Both sequential and reverse curricula result in dramatic degradation of first-stage knowledge, yet the simple-to-complex (sequential) ordering exhibits comparatively less forgetting than complex-to-simple (reverse), suggesting that the richer linguistic patterns learned from stories provide a more stable foundation that is harder to fully overwrite than the narrower news domain representations.
3. **Forgetting is asymmetric across domains.** News is forgotten at a disproportionately higher rate (37–95×) than stories (15–20×), driven by the narrow and formulaic nature of news text being more easily overwritten by the richer stories distribution.
4. **Training order determines where forgetting occurs in the network.** Sequential training (simple → complex) distributes weight changes uniformly across all layers, while reverse training (complex → simple) concentrates changes almost entirely in the `lm_head`, leaving deeper representational layers largely intact.
5. **Mixed training establishes a strong upper bound.** Joint training on both domains consistently outperforms both continual learning orderings across all model scales.
6. **Forward transfer is real but modest.** All models exhibit positive forward transfer (0.18–0.34× improvement), suggesting that prior domain exposure provides a useful initialisation, though the benefit is insufficient to compensate for the forgetting incurred.
7. **BERTScore masks forgetting that coverage scores expose.** Surface-level semantic quality remains stable across all conditions (~0.80 BERTScore), while coverage scores reveal a sharp shift in generative distribution toward the most recently trained domain.

Conclusion

This study examined catastrophic forgetting in small decoder-only Transformer LMs across three model scales and three training regimes. Results confirm that sequential domain training causes significant and measurable forgetting. M1 showed a normalised BWT of +18.16, meaning Domain A perplexity increased by over 1800% after Stage 2. Forgetting is asymmetric across domain order, and mixed training (E2) consistently outperforms both sequential regimes, establishing joint training as the upper bound. Scale effects reveal a tension between greater model capacity and worsening data-to-parameter ratio, whose resolution is captured in the experimental results. Semantic metrics via BERTScore and concept coverage supports PPL findings, while weight change analysis identifies FFN layers as the primary site of domain adaptation. Future work should explore mitigation strategies such as EWC, replay buffers, and larger domain corpora.

